

9.2 Resistance

Question Paper

Course	CIEA Level Physics
Section	9. Electricity
Topic	9.2 Resistance
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Which equation is used to define resistance?

- A. energy = (current)² × resistance × time
- B. potential difference = current × resistance
- C. power = (current)² × resistance
- D. resistivity = resistance × area ÷ length

[1 mark]

Question 2

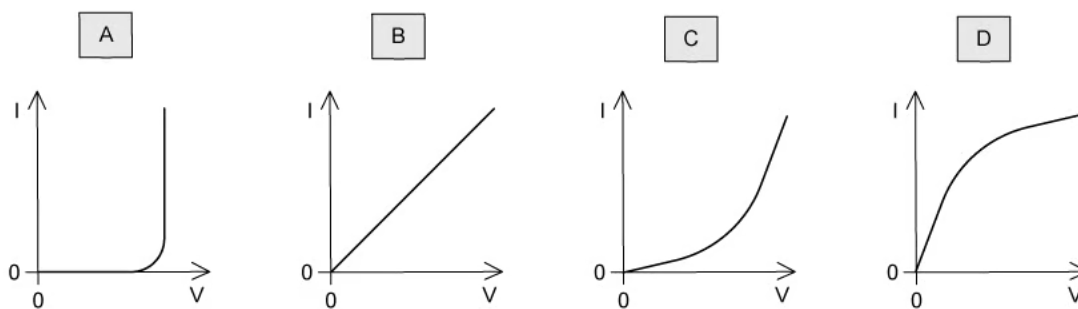
What is the unit of resistance when expressed in SI base units?

- A. $\text{kg m}^2 \text{s}^{-2} \text{A}^{-1}$
- B. $\text{kg m}^2 \text{s}^{-3} \text{A}^{-2}$
- C. $\text{kg m s}^{-2} \text{A}^{-1}$
- D. $\text{kg m s}^{-3} \text{A}^{-1}$

[1 mark]

Question 3

Which graph shows the I - V characteristic of a filament lamp?



[1 mark]

Question 4

The resistance R of a resistor is determined by measuring the potential difference V across it and the current I in it. The value of R is then calculated using the equation

$$R = \frac{V}{I}$$

The values measured are $V = 1.00 \pm 0.05 \text{ V}$ and $I = 0.50 \pm 0.01 \text{ A}$.

What is the percentage uncertainty in the value of R ?

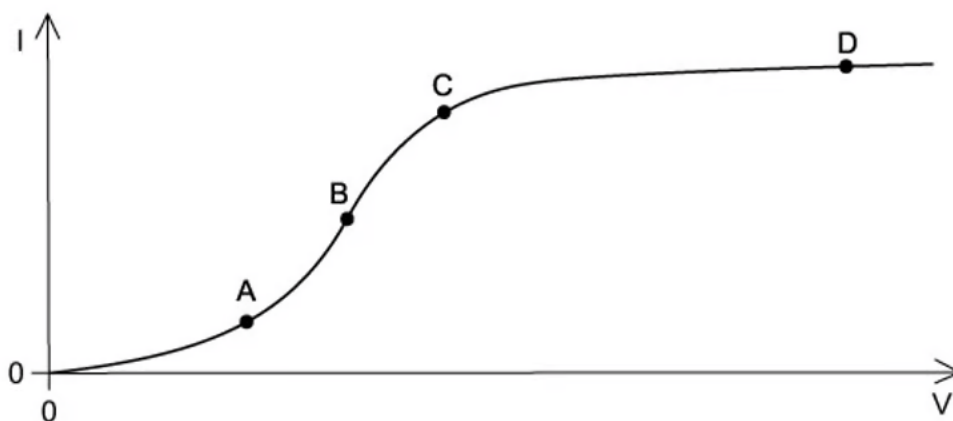
- A. 2.5 %
- B. 3.0 %
- C. 7.0 %
- D. 10.0 %

[1 mark]

Question 5

The graph shows how the electric current I through a conducting liquid varies with the potential difference V across it.

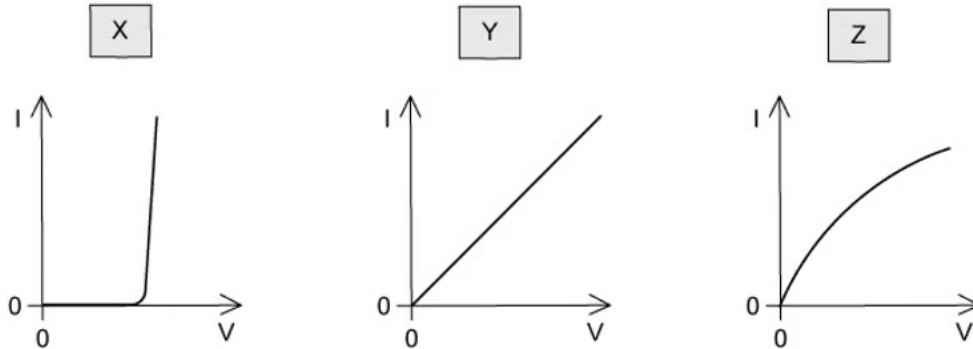
At which point on the graph does the liquid have the smallest resistance?



[1 mark]

Question 6

The graphs show the variation with potential difference V of the current I for three circuit components.



The components are a metal wire at constant temperature, a semiconductor diode and a filament lamp.

Which row of the table correctly identifies these graphs?

	Metal wire at constant temperature	Semiconductor diode	Filament lamp
A	X	Z	Y
B	Y	X	Z
C	Y	Z	X
D	Z	X	Y

[1 mark]

Question 7

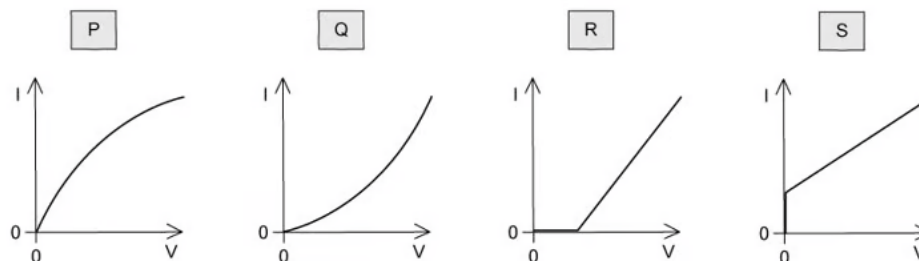
What is a correct statement of Ohm's Law?

- A. the potential difference across a component equals the current providing the resistance and other physical conditions stay constant
- B. the potential difference across a component equals the current multiplied by the resistance
- C. the potential difference across a component is proportional to its resistance.
- D. the potential difference across a component is proportional to the current in it providing physical conditions stay constant

[1 mark]

Question 8

The graphs show possible current–voltage (I – V) relationships for a filament lamp and a semiconductor diode.



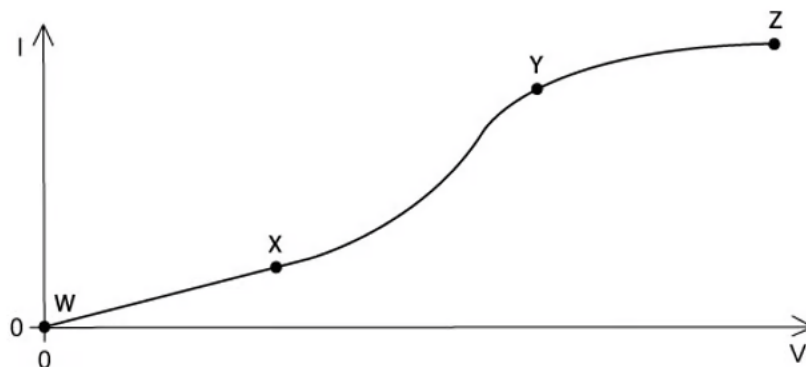
Which row best specifies the correct I – V graphs for the lamp and the diode?

	Filament lamp	Semiconductor diode
A	P	R
B	P	S
C	Q	R
D	Q	S

[1 mark]

Question 9

An electrical component has a potential difference V across it and a current I through it. A graph of I against V is drawn and is marked in three sections WX, XY and YZ.



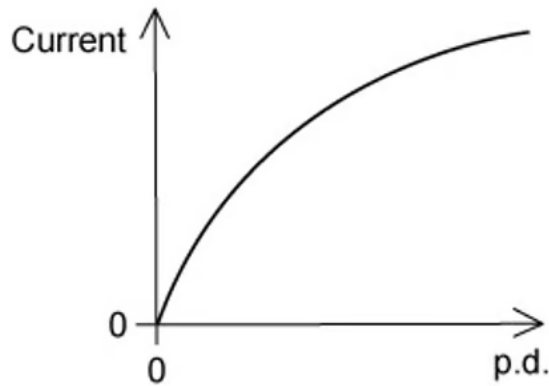
In which ways does the resistance of the component vary within each of the three sections?

	WX	XY	YZ
A	constant	decreases	increases
B	constant	increases	increases
C	increases	decreases	constant
D	increases	increases	decreases

[1 mark]

Question 10

The graph shows the variation with potential difference (p.d.) of the current in a lamp filament.



Which statement explains the shape of this graph?

- A. as the filament temperature rises, electrons can pass more easily through the filament
- B. it takes time for the filament to reach its working temperature
- C. the power output of the filament is proportional to the square of the current in it
- D. the resistance of the filament increases with a rise in temperature

[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- Resistance is defined by Ohm's Law, which states:
- The potential difference across a component is proportional to the current through it provided environmental conditions (e.g. temperature and pressure) stay constant
 - In other words, $V \propto I$
- From this proportionality, we can say $V = kI$, where k is a constant
- This constant is what is known as resistance R
 - So, $V = IR$ is the definition of resistance

2

The correct answer is **B** because:

- Resistance is defined as the ratio of voltage V across a component to the current I through it
 - In equation form, this can be expressed as $R = \frac{V}{I}$
- Potential difference is defined as the amount of electrical energy transferred between two points per unit of charge
 - In equation form, this can be expressed as $V = \frac{E}{Q}$
- The SI base units of potential difference are
 - $V = \frac{E}{Q} = \frac{\frac{1}{2}mv^2}{It} = [kg \ m^2 \ s^{-2}][A \ s]^{-1} = kg \ m^2 s^{-3} A^{-1}$
- So, the SI base units of resistance are
 - $R = \frac{V}{I} = [kg \ m^2 \ s^{-3} \ A^{-1}][A]^{-1} = kg \ m^2 \ s^{-3} A^{-2}$

3

The correct answer is **D** because:

- The resistance of a filament lamp increases with temperature so the ratio $\frac{V}{I}$ increases as current increases
- As a result, the gradient decreases and a greater change in the potential difference is required to cause a change in the current
- On the I - V graph, it can be seen as the gradient on the curve becomes shallower, indicating a greater resistance
- The reason for this is that as the potential difference across a filament lamp increases, the current increases and the energy dissipated, as heat, increases, resulting in a higher temperature
- As the temperature increases, the resistance of the filament increases and the number of collisions between the free electrons and the lattice ions increases due to more rigorous vibration of the lattice ions

A is incorrect because this is the I - V graph for a semiconductor diode

B is incorrect because this is the I - V graph for an Ohmic conductor (obeys Ohm's Law)

C is incorrect because this is the I - V graph for a thermistor

4

The correct answer is **C** because:

- From the equation given, the formula to work out the uncertainty in resistance is
 - $\left(\frac{\Delta R}{R}\right) = \left(\frac{\Delta V}{V}\right) + \left(\frac{\Delta I}{I}\right)$
- To work out uncertainty in data which is multiplied or divided we add the percentage uncertainties
 - $\% \text{ uncertainty} = \frac{\text{uncertainty}}{\text{value}} \times 100$
 - $\left(\frac{\Delta R}{R} \times 100\right) = \left(\frac{0.05}{1.00} \times 100\right) + \left(\frac{0.01}{0.50} \times 100\right) = 5.0 \% + 2.0 \% = 7.0 \%$

5

The correct answer is **B** because:

- Ohm's Law is defined by the equation $V = IR$
- Resistance, however, is the potential difference per unit current $R = \frac{V}{I}$

- In relation to the I - V graph, we are looking for the point on the curve which gives us the maximum gradient
 - This is because the gradient is equal to $\frac{\Delta y}{\Delta x} = \frac{I}{V} = \frac{1}{R}$ so a maximum gradient will give us the smallest R
- The point with the maximum gradient is point **B**

6

The correct answer is **B** because:

- Graph X describes a conductor which has a very low resistance in one direction (low $\frac{V}{I}$) and a very high resistance in the other direction (high $\frac{V}{I}$)
 - This matches the description of a semiconductor diode
- Graph Y describes a conductor which obeys Ohm's Law, which means:
- The potential difference across a component is proportional to the current through it provided physical conditions (e.g. temperature and pressure) stay constant – in other words, $V \propto I$
 - This matches the description of a metal wire at constant temperature
- Graph Z describes a conductor in which the resistance increases due to a change in environmental conditions (e.g. temperature) as the ratio $\frac{V}{I}$ increases
 - This matches the description of a filament lamp

7

The correct answer is **D** because:

- Resistance is defined by Ohm's Law, which states:
- The potential difference across a component is proportional to the current through it provided physical conditions (e.g. temperature and pressure) stay constant
 - In other words, $V \propto I$
- From this proportionality, we can say $V = kI$, where k is a constant
- This constant is what is known as resistance R
 - So, $V = IR$ is the definition of resistance

8

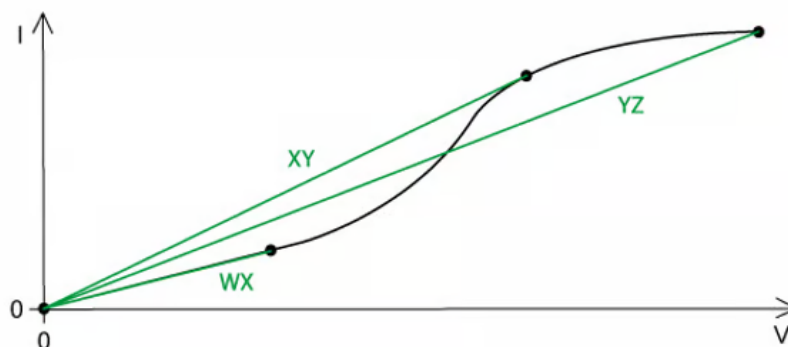
The correct answer is **A** because:

- Graph P describes a conductor in which the resistance increases due to a change in environmental conditions (e.g. temperature) as the ratio $\frac{V}{I}$ increases
 - This matches the description of a filament lamp
- Graph R describes a conductor which has a very low resistance in one direction (low $\frac{V}{I}$) and a very high resistance in the other direction (high $\frac{V}{I}$)
 - This matches the description of a semiconductor diode

9

The correct answer is **A** because:

- Resistance is defined as the potential difference per unit current $R = \frac{V}{I}$
- In section WX, the straight line indicates $V \propto I$, so R is constant here
- In section XY, the current is increasing at a higher rate than the potential difference, which leads to a decrease in R
 - As well as this, the gradient becomes steeper, meaning $\frac{I}{V} = \frac{1}{R}$ is increasing, which leads to a decrease in R
- In section YZ, the potential difference is increasing at a higher rate than the current, which leads to an increase in R
 - As well as this, the gradient becomes less steep, meaning $\frac{I}{V} = \frac{1}{R}$ is decreasing, which leads to an increase in R



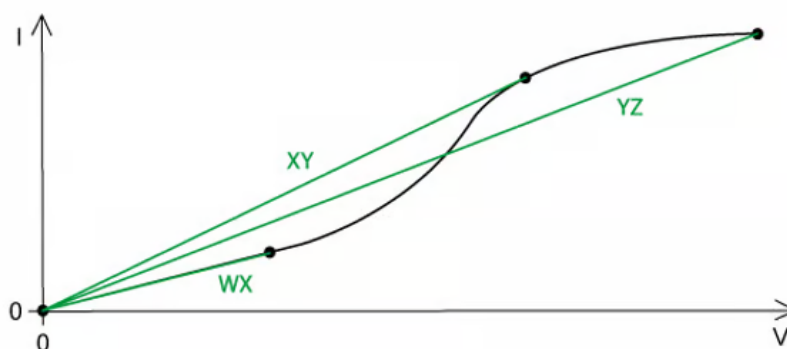
10

The correct answer is **D** because:

- Ohm's Law is defined by the equation $V = IR$
- At any point on the graph, the ratio of $\frac{I}{V}$ gives the value of $\frac{1}{R}$ where R is the resistance of the filament lamp
- It can be seen from the graph that this ratio decreases as the p. d. increases

The correct answer is **A** because:

- Resistance is defined as the potential difference per unit current $R = \frac{V}{I}$
- In section WX, the straight line indicates $V \propto I$, so R is constant here
- In section XY, the current is increasing at a higher rate than the potential difference, which leads to a decrease in R
 - As well as this, the gradient becomes steeper, meaning $\frac{I}{V} = \frac{1}{R}$ is increasing, which leads to a decrease in R
- In section YZ, the potential difference is increasing at a higher rate than the current, which leads to an increase in R
 - As well as this, the gradient becomes less steep, meaning $\frac{I}{V} = \frac{1}{R}$ is decreasing, which leads to an increase in R



10

The correct answer is **D** because:

- Ohm's Law is defined by the equation $V = IR$
- At any point on the graph, the ratio of $\frac{I}{V}$ gives the value of $\frac{1}{R}$ where R is the resistance of the filament lamp
- It can be seen from the graph that this ratio decreases as the p. d. increases since the gradient decreases
- So, this means that the resistance R of the filament lamp increases
- When resistance increases, less current can flow
- Note that when current passes through the filament, its temperature increases

- Therefore, from the graph, it can be concluded that the resistance of the filament increases with a rise in temperature

A is incorrect because as the filament temperature rises, the resistance increases, so electrons pass through the filament **less** easily

B & C are incorrect because these options do not explain the shape of the graph

